costsight

Automated Cloud Cost Anomaly Detection

Project 13 · Cloud Computing · Spring 2025–2026

Furkan Can Karafil · Halil Utku Demirtaş

https://github.com/Urthella/costsight

The Problem

> 30%+ of cloud spend is wasted — *Flexera State of the Cloud, 2024*

The damage is already done by the time it shows up on the bill:

- **Runaway autoscaling** — misconfigured rules trigger resource explosions
- **Software bugs & leaks** — infinite loops can inflate the bill within hours
- **Invisible resources** — forgotten test environments, untagged resources

Cost anomalies need to be detected in **hours**, not weeks.

Our Solution — End-to-End Pipeline

```
Synthetic AWS CUR  →  Preprocessing  →  Three detectors  →  Alerts  →  Dashboard
                                        • Z-Score
                                        • STL Decomposition          severity =
                                        • Isolation Forest           deviation
                                                                     × duration
                                                                     × $impact
```

Outcome: a practical FinOps tool that catches anomalies in hours, with full transparency about *which* algorithm caught *which* anomaly.

What Sets This Apart

	costsight	AWS Cost Explorer	Commercial tools
Detection algorithms	**3 compared**	1 closed	1 black box
Per-anomaly-type metrics	**Yes**	No	No
Severity scoring	**Open formula**	Limited	Proprietary
Reproducibility	**MIT, GitHub**	Closed-source	Closed

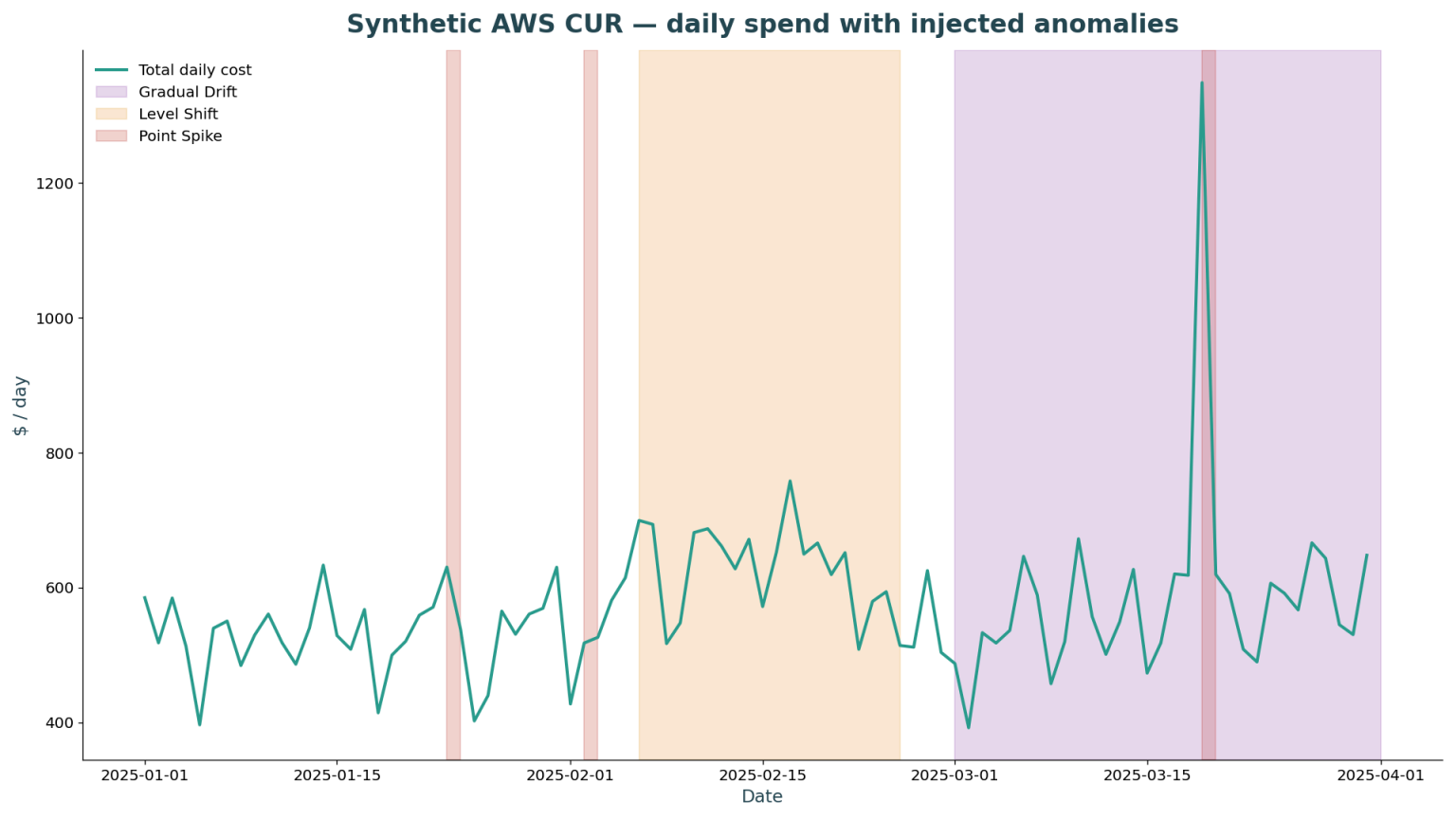
> **Gap we address:** cloud-cost-specific comparison across anomaly types remains limited in open literature.

Synthetic Data — A Sneak Peek

Three anomaly types are injected with **ground-truth labels** so we can compute Precision/Recall:

- **Point spike** — single-day cost explosion
- **Level shift** — persistent step up
- **Gradual drift** — slow upward creep

90 days × 7 AWS services ×



Three Detection Methods

Method	Type	Strengths
Z-Score	Statistical baseline	Fast, interpretable, perfect on point spikes
STL	Time-series decomposition	Handles seasonality, drift, level shift
Isolation Forest	ML / ensemble	Multi-feature anomalies, no labels needed

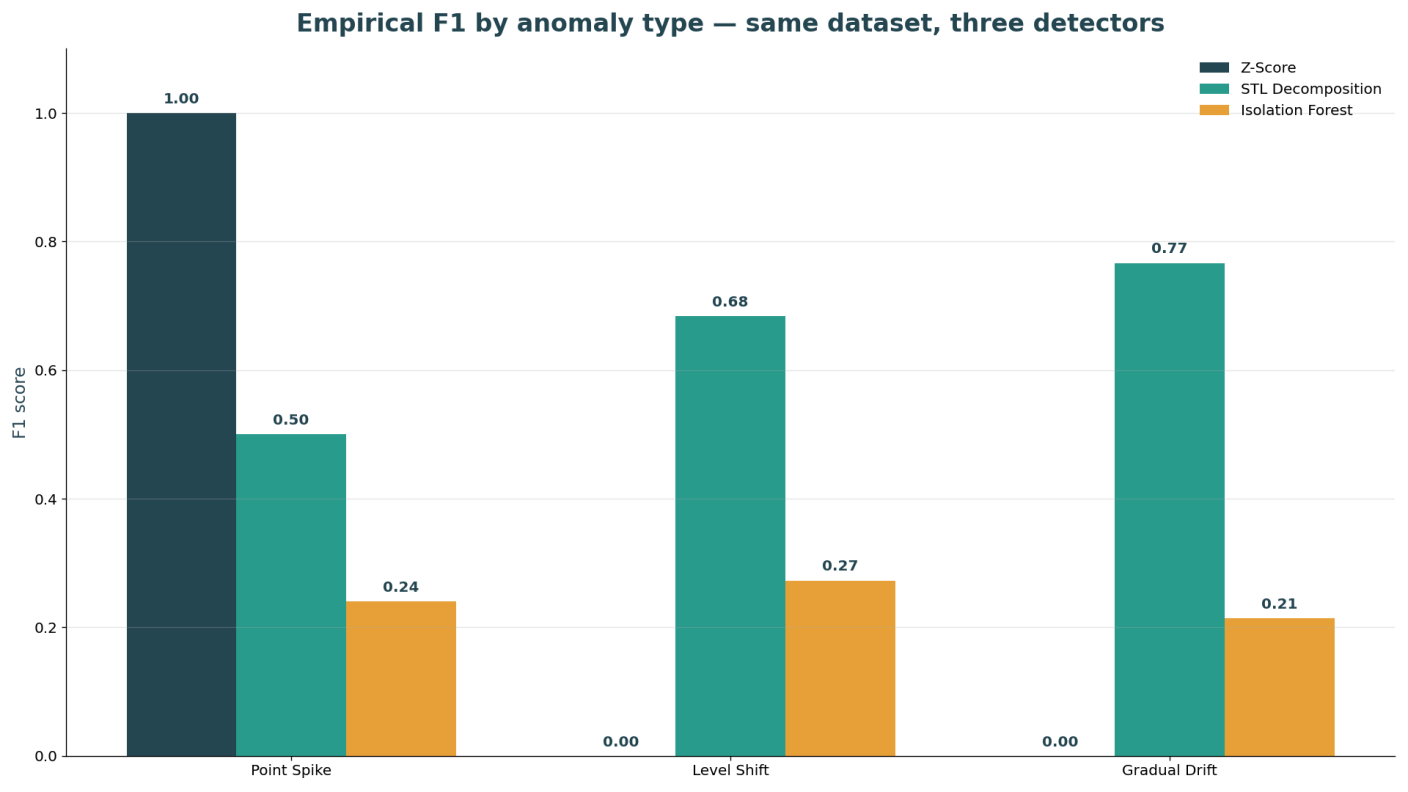
Each detector exposes the same `detect(df)` interface — alerts and evaluation are **detector-agnostic**.

Empirical F1 by Anomaly Type

Three observations:

- Z-Score is **perfect** on point spikes
- Z-Score is **blind** to drift / level shift
- STL is **strongest overall**
- Isolation Forest is **mid-pack** but consistent across types

> Same data, three lenses — measured, not estimated.

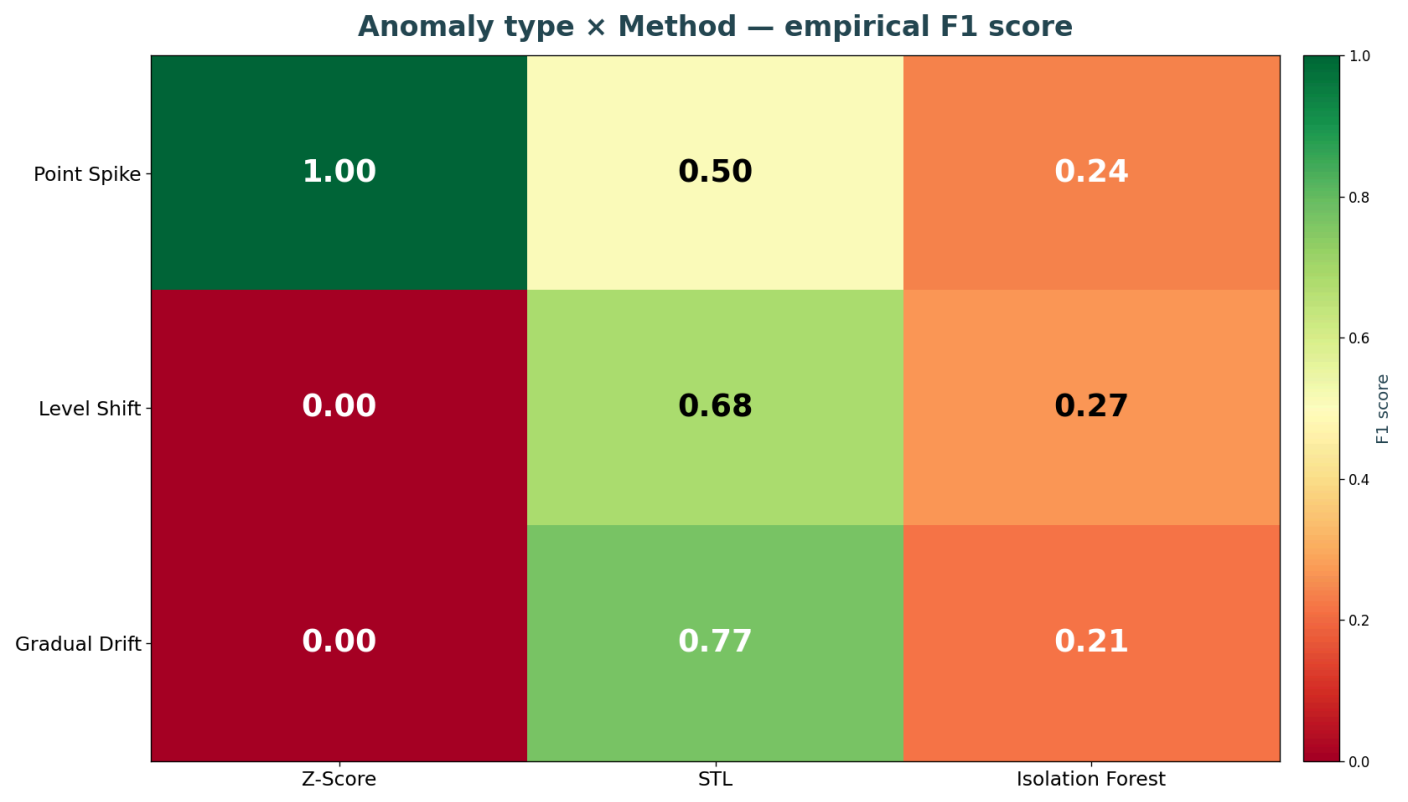


Anomaly Type × Method Performance

Empirical F1 across 25 random seeds, mean values:

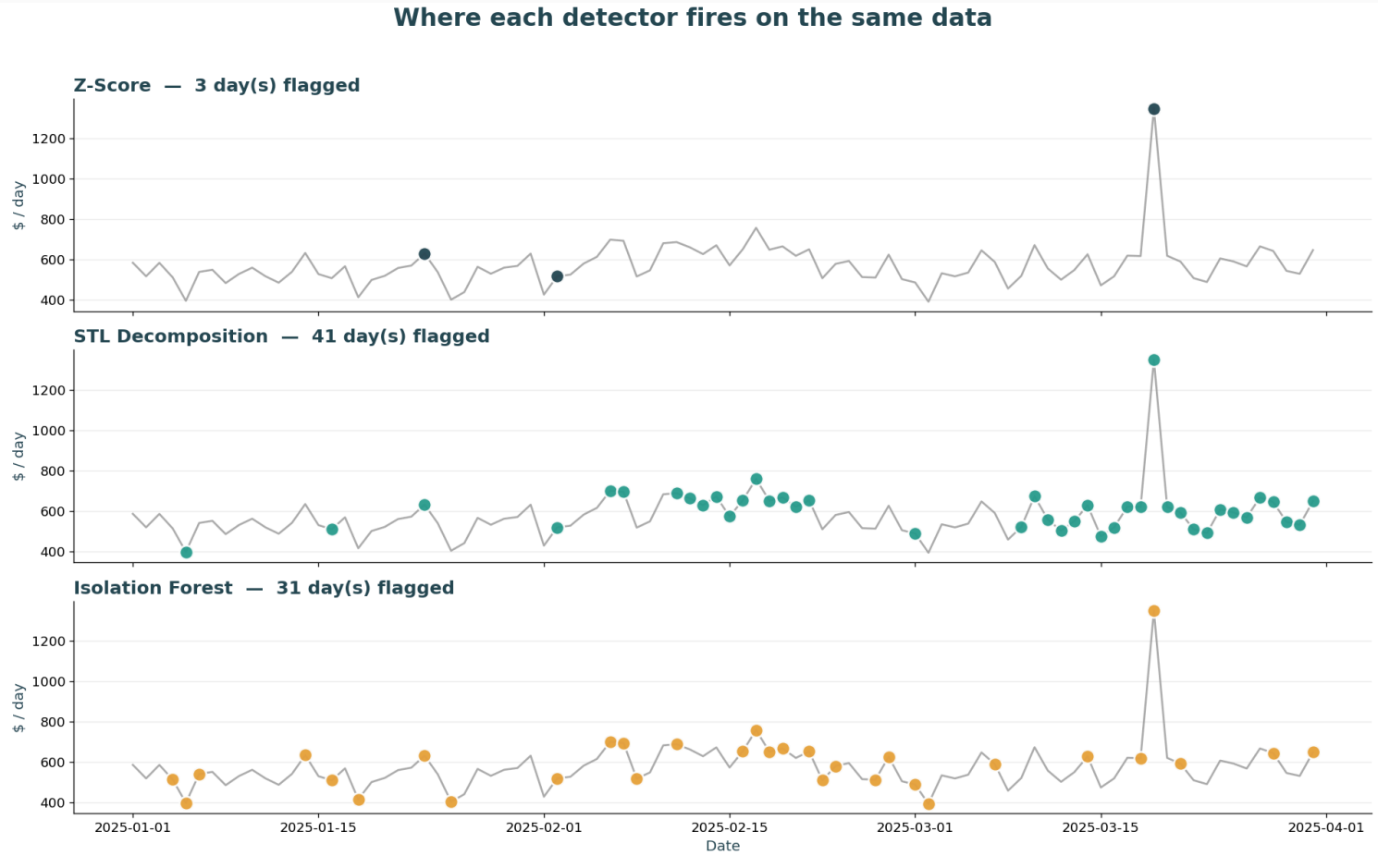
- Z-Score → 0.96 / 0.01 / 0.00
- STL → 0.52 / 0.62 / 0.73
- iForest → 0.25 / 0.22 / 0.22

Takeaway: no single method wins all types — that's why we run all three.



Where Each Detector Fires

- Z-Score lights up only the obvious spikes (3 days)
- STL captures spikes + drift + level shifts (~41 days)
- Isolation Forest adds a few mid-window flags (~31 days)



Multi-Seed Robustness

25 independent random seeds, mean ± std F1:

Detector	Point Spike	Level Shift	Gradual Drift	
Z-Score	0.962 ± 0.078	0.012 ± 0.033	0.000 ± 0.000	**0.105 ± 0.018**
STL	0.522 ± 0.082	0.616 ± 0.204	0.734 ± 0.052	**0.757 ± 0.064**
Isolation Forest	0.247 ± 0.035	0.216 ± 0.060	0.217 ± 0.034	**0.319 ± 0.036**

> Std is tight: STL's lead is **statistically robust**, not a single-seed fluke.

Alert Quality by Severity

The severity formula `deviation × duration × $impact` is a triage filter.

Severity	STL precision	iForest precision	Z-Score precision
MEDIUM	1.000	1.000	1.000
LOW	0.860	0.406	1.000

MEDIUM and HIGH alerts are ~100% true positives across detectors — a FinOps engineer who only triages MEDIUM+ sees almost no false alarms.

Tech Stack

- **Python 3.11+** — language
- **pandas / NumPy / PyArrow** — data processing
- **statsmodels** — STL Decomposition
- **scikit-learn** — Isolation Forest
- **SciPy** — statistics primitives
- **Streamlit / Plotly / Matplotlib** — dashboard + figures
- **GitHub Actions** — CI on Python 3.11 and 3.12

> One install (`pip install -r requirements.txt`), one command (`python scripts/run_pipeline.py`), every result reproducible.

Achieved Deliverables

- **Working Python pipeline** — 3 detectors, all green on CI
- **Automated alert system** — JSON + CSV, severity-banded
- **Root-cause hints** — "*us-east-1 region drove 100% of the increase*"
- **Live Streamlit dashboard** — 5 tabs incl. root-cause + comparison
- **Documented GitHub repo** — README, REPORT, DEMO, examples, MIT license
- **Multi-seed benchmark** — 25 seeds, mean ± std reported
- **Demo video** — 2-minute walkthrough of the dashboard

Limitations & Future Work

Limitation	Future direction
Synthetic data only	Onboard anonymized real CUR exports
Univariate per service	Multi-feature & multi-granularity (account/region/tag)
Batch-only execution	Streaming via Kafka / Kinesis
Heuristic severity formula	Learn band thresholds from FinOps feedback
Single cloud (AWS)	Multi-cloud schema (GCP, Azure)

> Phase 1 scope is intentionally bounded — these are Phase 2 / Level 2 work.

Work Division

Phase 1 / Phase 2 — Joint Contributions

- **Halil Utku Demirtaş** — CUR generator, preprocessing, alert module, Precision / Recall framework, system architecture
- **Furkan Can Karafil** — STL implementation, Isolation Forest model, Z-Score baseline, Streamlit dashboard, comparative analysis
- **Joint** — GitHub repo, CI, multi-seed benchmark, technical report, slide deck, demo video

Thank you

We welcome your questions

Repository: https://github.com/Urthella/costsight

Demo video: see `DEMO.md` for the recorded walkthrough

Report: `REPORT.md` for the full technical write-up

Furkan Can Karafil · Halil Utku Demirtaş